LEUKEMIA

Four types. Four memory scenes.

A companion to your four original illustrated cards and the Simple Nursing handout. These are the four main types; rarer leukemias are outside this overview.

01 Sort by speed + cell lineage

DISEASE COURSE	LYMPHOID	MYELOID
ACUTE Rapid; immature cells	**ALL** Lymphoblast express	**AML** Marrow mine
CHRONIC Usually slower at first	**CLL** Slow B-cell library	**CML** Philadelphia factory

02 Shared nursing priorities

FEVER DURING CHEMOTHERAPY

38°C (100.4°F) or higher

Contact oncology immediately. Follow the facility neutropenic-fever pathway. High total WBCs do not prove effective infection protection. [5, 6]

TUMOR LYSIS RISK

Watch especially with high disease burden or treatment initiation. Trend **potassium, phosphate, calcium, uric acid, creatinine and urine output.** Provide ordered hydration and prophylaxis. [7]

K / phosphate / uric acid rise — **Calcium falls**

Corrections and clickable sources on page 2

SOURCE COMPANION

Read the handout with these corrections

Keep these distinctions beside the original Simple Nursing page.

AGE: ALL IS THE CHILDHOOD LEADER

The pediatric note beside AML is misleading. **ALL is the most common childhood leukemia.** AML becomes more common with age. Both can occur in children and adults. [1, 2]

LABS: COUNTS CAN VARY

WBCs may be high, normal or low, especially in acute leukemia. Platelets are not always low; they can be elevated in CML. Use the differential, marrow/flow testing and genetics as indicated. [5]

TREATMENT: MATCH THE SUBTYPE

Radiation plus chemotherapy is not a universal plan. **CML usually uses a TKI; some stable CLL is monitored.** Acute leukemia needs prompt specialist treatment; **suspected APL is an emergency.** [1-4, 8]

Sources & attribution

The four memory cards are original AI-generated illustrations, checked against these clinical sources. They are not official Picmonic or Simple Nursing artwork. The original Simple Nursing handout is preserved on page 7. Separate subtype-specific Simple Nursing handouts were not verified.

1 NCI: ALL

2 NCI: AML

3 NCI: CLL

4 NCI: CML

5 Merck: leukemia comparison

6 CDC: infection and fever

7 Canadian Cancer Society: TLS

8 ELN: suspected APL urgency

Simple Nursing: public leukemia lesson

Original Simple Nursing source PDF

ALL

ACUTE LYMPHOBLASTIC LEUKEMIA

ALL aboard the lymphoblast express!



1 Fast train → ACUTE: rapid progression

2 Young conductor → Most common childhood leukemia; adults get it too

3 Baby lymphocytes → Immature B- or T-cell precursors

4 Crowded marrow → Anemia, infection, bleeding

5 Bone station → Bone or joint pain may occur

6 Brain shield → CNS prophylaxis is part of treatment

TREATMENT SNAPSHOT

Multi-drug therapy in phases; maintenance is often included. Targeted or immune therapy depends on subtype.

PRIORITY: FEVER DURING CHEMOTHERAPY

Temperature ≥38°C (100.4°F): notify the oncology team immediately. Follow the neutropenic-fever protocol.

Clinical basis: *NCI ALL summaries; CDC infection guidance.*

Original memory visual • Nursing study aid • 2026

AML

ACUTE MYELOID LEUKEMIA

The marrow mine is blasting out of control!



1. **Blasting mine** → ACUTE myeloid precursor cancer
2. **Older miner** → More common in older adults; any age possible
3. **Auer rods** → May appear in AML blasts; not present in every case
4. **Empty bins** → Fatigue, infections, bruising or bleeding
5. **APL alarm** → Acute promyelocytic leukemia: high risk of DIC



TREATMENT SNAPSHOT

- Chemotherapy and/or targeted therapy; transplant for selected patients.
- APL has distinct treatment: ATRA-based therapy, often with arsenic trioxide.

PRIORITY: SUSPECTED APL IS AN EMERGENCY

Escalate immediately. Monitor bleeding and coagulation; prepare urgent ordered ATRA and blood-product support.

DIC = disseminated intravascular coagulation | *Clinical basis: NCI AML summaries; Merck Manual.*

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CLL

CHRONIC LYMPHOCYTIC LEUKEMIA

The slow B-cell library: watch, count, protect.



TREATMENT SNAPSHOT

Stable and asymptomatic: active surveillance may be appropriate.

Progressive or symptomatic: targeted therapy, often BTK- or BCL-2-directed.

Watching means scheduled monitoring—not ignoring the disease.

NURSING FOCUS: PROTECT + MONITOR

Report fever promptly. Track CBC, infections, fatigue, node changes and follow-up visits.

Clinical basis: NCI CLL summary; Merck Manual. | *Original memory visual • Nursing study aid* • 2026

CML

CHRONIC MYELOID LEUKEMIA

Philadelphia's myeloid factory has a stuck ON switch.



5 Blast rocket → Can progress to an aggressive blast phase

4 Spleen balloon → Splenomegaly; left upper abdominal fullness

3 Myeloid workers → Excess granulocytes; often high WBCs

1 Stuck switch → BCR::ABL1 drives uncontrolled growth

2 9 ↔ 22 swap → Philadelphia chromosome, t(9;22)

TREATMENT SNAPSHOT

TKI

- Usually an oral tyrosine kinase inhibitor (TKI), such as imatinib.
- Monitor CBC and BCR::ABL1 molecular response.

Chronic does not mean harmless or untreated.

NURSING FOCUS: ADHERENCE + FOLLOW-UP

Take TKI exactly as prescribed. Review interactions and side effects. Keep blood-test and molecular monitoring visits.

Clinical basis: NCI CML summary. | Original memory visual • Nursing study aid • 2026

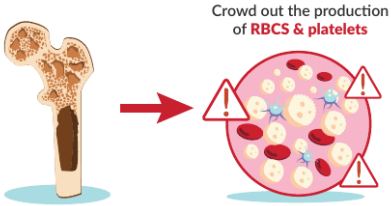
Leukemia

Med Surg: Oncology (Cancer)

SimpleNursing

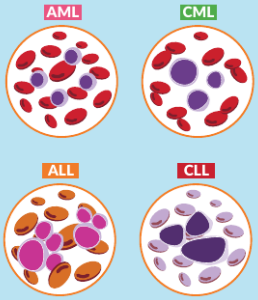
Pathophysiology

Leukemia is a type of cancer that affects the **blood cells & bone marrow**. Bone marrow is responsible for making blood cells: WBCs, RBCs, platelets. In Leukemia there is an **overproduction of white blood cells (WBCs)** that crowd out the production of normal cells, leading to low RBCs & low platelets.



Leukemia
- Leuk = Leukocyte (WBC)
- emia = blood

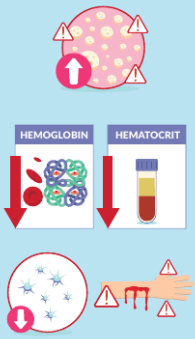
Main Types



- **AML** - Acute myeloid Leukemia
  Pediatrics more common
- **CML** - Chronic myeloid Leukemia
- **ALL** - Acute Lymphocytic Leukemia
- **CLL** - Chronic Lymphocytic Leukemia

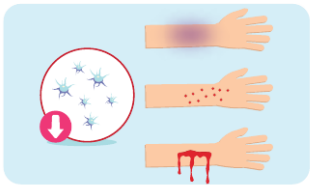
Labs

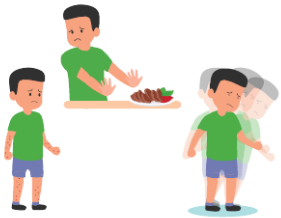
Labs
- High WBCs
- Low H/H
- Low Platelets



Signs & Symptoms

- **Frequent infections**
- **Fatigue**, Unsteady Gait, Pale "Pallor"
- **Bruising**, Petechiae, & easy Bleeding
- Weight Loss & **Anorexia**
- Bone pain







ATI Question

A nurse is **assessing** a child who has **leukemia**. Which of the following **findings should the nurse expect**?
Select all that apply.

- Anorexia
- Petechiae on the extremities
- Unsteady gait

ATI Question

7-year-old with acute lymphoblastic leukemia who is being admitted for **evaluation of fever**. To which of the **following rooms** should the nurse assign the client?

- **Private room** in order to **prevent transmission of infection** from others

Diagnostics

Bone marrow biopsy
Taken from the posterior iliac crest



HESI Question

A nurse preparing a patient for a **bone marrow biopsy** knows **which site** will most likely be used?

- **Posterior iliac crest**

Treatment

Radiation & Chemotherapy to kill the cancer.
Rarely: Stem cell transplant - like hitting the restart button to reboot the bone marrow.



